

~\OneDrive\Dokument\SKOLA\kiruna\Arne\Arduino\Arne\Arne.ino

```
1  /* Arne.ino
2  * Version: 1.4
3  * Created: 2023-04-21
4  * Author: Hjalmar Stenberg
5  *
6  * Purpose: Reads sensor data and writes it to SD card
7  * System: ESP32 Feather V2
8  * Uses: Mapf.h, ADC_Cal.h, SD_Lib.h, MPU6050_Lib.h
9  *****/
10
11 #include <Mapf.h>
12 #include "ADC_Cal.h"
13 #include "SD_Lib.h"
14 #include "MPU6050_Lib.h"
15
16 #define ACC_SCALE 200
17 #define MAX_VALUE_ADC 4095
18 #define PRINT_SERIAL 0
19 #define WIPE_ON_RESET 1
20 #define IS_TESTBENCH 0
21
22 //Arrays for ADC Calibration look-up tables
23 float ADC1_LUT[4096];
24 float ADC2_LUT[4096];
25
26 //Defining variables for error correction
27 float Error_X;
28 float Error_Y;
29 float Error_Z;
30
31 float Error_X200g;
32 float Error_Y200g;
33 float Error_Z200g;
34
35 float Error_Gyro_X;
36 float Error_Gyro_Y;
37 float Error_Gyro_Z;
38
39 void setup() {
40     //Testbench-board has different errors and look-up tables
41     if(IS_TESTBENCH){
42         memcpy(ADC1_LUT, ADC1_TB, sizeof(ADC1_TB[0])*4096);
43         Error_X = -0.6033;
44         Error_Y = 0.7590;
45         Error_Z = -0.9998;
46
47         Error_X200g = -0.42;
48         Error_Y200g = -1.2015;
49         Error_Z200g = 1.9084;
50
51         Error_Gyro_X = 0;
52         Error_Gyro_Y = 0;
53         Error_Gyro_Z = 0;
54     }
55     else {
56         memcpy(ADC1_LUT, ADC1_KiB, sizeof(ADC1_KiB[0])*4096);
```

```

57     Error_X = 0.4429;
58     Error_Y = 6.0238;
59     Error_Z = 1.3996;
60
61     Error_X200g = 3.7216;
62     Error_Y200g = 2.6667;
63     Error_Z200g = 1.1978;
64
65     Error_Gyro_X = 0;
66     Error_Gyro_Y = 0;
67     Error_Gyro_Z = 0;
68 }
69
70 //Initialize I2C communication
71 Wire.begin(SDA,SCL,400000);
72
73 //Initialize ADC
74 analogReadResolution(12);
75 Serial.begin(115200);
76
77 //Initialize SD card reading
78 initSD(SD,WIPE_ON_RESET);
79
80 //Initialize MPU6050
81 initMPU6050();
82 }
83
84
85 //Opens data-file, writes 5000 readings, then closes
86 void loop() {
87     openArne(SD);
88     for(int i=0;i<=5000;i++){
89         //Get accelerometer readings
90         float rawX = ADC1_LUT[analogRead(A2)];
91         float rawY = ADC1_LUT[analogRead(A3)];
92         float rawZ = ADC1_LUT[analogRead(A4)];
93
94         //Mapping raw data to g-forces
95         float scaledX = mapf(rawX, 0, MAX_VALUE_ADC, -ACC_SCALE, ACC_SCALE);
96         float scaledY = mapf(rawY, 0, MAX_VALUE_ADC, -ACC_SCALE, ACC_SCALE);
97         float scaledZ = mapf(rawZ, 0, MAX_VALUE_ADC, -ACC_SCALE, ACC_SCALE);
98
99         //Fixing errors
100        scaledX -= Error_X200g;
101        scaledY -= Error_Y200g;
102        scaledZ -= Error_Z200g;
103
104        //Calculating total acceleration
105        double accTot = sqrt(pow(scaledX,2) + pow(scaledY,2) + pow(scaledZ,2));
106
107        //Make 200g acc readings to string
108        char scaledXStr[16];
109        dtostrf(scaledX,8,4,scaledXStr);
110        char scaledYStr[16];
111        dtostrf(scaledY,8,4,scaledYStr);
112        char scaledZStr[16];
113        dtostrf(scaledZ,8,4,scaledZStr);
114        char accTotStr[16];
115        dtostrf(accTot,8,4,accTotStr);
116

```

```

117 //Get good acc readings
118 sensors_event_t a, g, temp;
119 getReadingsMPU6050(&a,&g,&temp);
120
121 //Make good acc readings to string
122 char goodAccXStr[16];
123 dtostrf(a.acceleration.x-Error_X,8,4,goodAccXStr);
124 char goodAccYStr[16];
125 dtostrf(a.acceleration.y-Error_Y,8,4,goodAccYStr);
126 char goodAccZStr[16];
127 dtostrf(a.acceleration.z-Error_Z,8,4,goodAccZStr);
128
129 //Make gyroscope readings to string
130 char gyroX[16];
131 dtostrf(g.gyro.x-Error_Gyro_X,8,4,gyroX);
132 char gyroY[16];
133 dtostrf(g.gyro.y-Error_Gyro_Y,8,4,gyroY);
134 char gyroZ[16];
135 dtostrf(g.gyro.z-Error_Gyro_Z,8,4,gyroZ);
136
137 //Get timestamp
138 int currentTime = micros();
139 char currentTimeStr[16];
140 itoa(currentTime,currentTimeStr,10);
141
142 //Write to file
143 printArne(scaledXStr);
144 printArne(",");
145 printArne(scaledYStr);
146 printArne(",");
147 printArne(scaledZStr);
148 printArne(",");
149 printArne(accTotStr);
150 printArne(",");
151 printArne(goodAccXStr);
152 printArne(",");
153 printArne(goodAccYStr);
154 printArne(",");
155 printArne(goodAccZStr);
156 printArne(",");
157 printArne(currentTimeStr);
158 printArne("\n");
159
160 //Printing to serial (mostly for testing purposes)
161 if(PRINT_SERIAL){
162 // Serial.print("Constant:");
163 // Serial.print(1);
164 // Serial.print(",");
165 Serial.print("X:");
166 Serial.print(goodAccXStr);
167 Serial.print(",");
168 Serial.print("Y:");
169 Serial.print(goodAccYStr);
170 Serial.print(",");
171 Serial.print("Z:");
172 Serial.println(goodAccZStr);
173
174 // Serial.print(",");
175 // Serial.print("GyroX:");
176 // Serial.print(gyroX);

```

```
177 //      Serial.print(",");
178 //      Serial.print("GyroY:");
179 //      Serial.print(gyroY);
180 //      Serial.print(",");
181 //      Serial.print("GyroZ:");
182 //      Serial.println(gyroZ);
183     }
184 }
185 closeArne();
186 }
187
```